Software Requirements Specification for Digital Health Record Management System (DHRMS)

Version 2.0 approved

Course: Software Engineering (SWE) • Academic Session 2026-2027

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Certificate of Recommendation and Approval

This is to certify that this Software Requirements Specification (SRS) document entitled "Digital Health Record Management System (DHRMS)", prepared and submitted by Abhinav Singh (Roll No: 202401100200014), Abhinav Pathak (Roll No: 202401100200013), Abhishek Kumar Verma (Roll No: 202401100200015), and Abhishek Yadav (Roll No: 202401100200016) in partial fulfillment of the academic requirements for the course Software Engineering in the Department of Computer Science and Engineering, is an authentic record of software engineering requirements analysis and specification work carried out under my supervision.

The functional requirements, interface specifications, performance constraints, and security controls specified in this document conform to the ISO/IEC/IEEE 29148:2018 standard and the structural guidelines of IEEE 830-1998. The specification has been reviewed and approved as the technical baseline for subsequent system architecture and development phases.

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Student Declaration of Originality

We hereby declare that this Software Requirements Specification document represents our authentic, original engineering work. All literature, regulatory standards (ISO/IEC/IEEE 29148:2018, HL7 FHIR Release 5, OWASP ASVS 5.0, HIPAA), and external technical specifications utilized during requirements elicitation have been duly acknowledged and formally cited in Section 1.5.

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Revision History

Name / Team	Date	Reason For Changes	Version
Project Team	2026-09-02	Initial project compilation of Section 1 (Introduction and Scope)	1.0
Project Team	2026-10-08	Draft functional requirements (4.1-4.10) and non-functional baseline	1.1
A. Singh, A. Pathak,A. K. Verma, A. Yadav	2026-10-10	Official Academic Release: Full ISO/IEC/IEEE 29148:2018 alignment, formal team attribution, faculty guide integration, complete textual analysis models, resolution of all 15 TBD decisions, removal of search token artifacts, verified academic citations [1]-[5]	2.0

1. 1. Introduction

1.1 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to establish a complete, formal, and verifiable technical baseline for the Digital Health Record Management System (DHRMS), Version 2.0. This document defines the system's functional requirements, behavioral characteristics, operational constraints, external hardware and software interfaces, security controls, and non-functional quality attributes.

This SRS serves as the primary technical baseline for system architects, developers, testers, academic evaluators, security reviewers, and project supervisor Mr. Navdesh Singh. The requirements are written so that each major capability can be implemented and verified independently.

1.2 1.2 Document Conventions

This document follows the requirements-engineering principles of ISO/IEC/IEEE 29148:2018 [1] and the structural conventions of IEEE 830-1998. The standard specifies requirements-engineering processes, information items, their content, and guidance for their format:

- System components and entity names appear in boldface (e.g., Patient, Healthcare Provider).
- Interface names, schema types, protocol tags, and technical identifiers appear in Courier-style notation where applicable (e.g., REQ-AUTH-001, application/json).
- Every individual functional requirement is assigned a unique identifier structured as REQ-[SUBSYSTEM]-[001-999].
- SHALL indicates a mandatory requirement; SHOULD indicates a recommended capability; MAY indicates an optional feature (in accordance with RFC 2119).
- High, Medium, and Low identify feature priority.

1.3 1.3 Intended Audience and Reading Suggestions

This document is intended for academic evaluators, software architects, system analysts, developers, testers, security reviewers, database designers, healthcare administrators, and future implementation teams.

Readers should begin with Section 1 to understand the purpose and scope, continue to Section 2 for the system context, Section 3 for external interfaces, Section 4 for detailed functional requirements, Section 5 for quality, performance, safety, security, and business rules, Section 6 for operational requirements, and Appendices A through D for detailed analysis models and engineering specifications.

1.4 1.4 Product Scope

The Digital Health Record Management System (DHRMS) is a centralized, secure web- and service-based software platform designed to manage the lifecycle of patient electronic health records, diagnostic reports, treatment histories, prescriptions, clinical encounters, consent records, and audit information.

System Objectives:

- Eliminate paper-dependent record handling by digitizing longitudinal patient histories into unified health profiles.
- Enforce granular consent mechanisms so that patients can authorize or revoke provider access to specified health information.
- Provide compliance-ready mechanisms for confidentiality, access control, integrity, and auditability.
- Facilitate interoperable healthcare data exchange using standardized healthcare data representations such as HL7 FHIR Release 5 [2].

Core Benefits:

- Patients receive centralized visibility into their health information and control over applicable provider access.
- Healthcare providers receive authenticated access to authorized clinical histories and can maintain clinical information in a structured manner.
- Enterprise administrators receive role-based permission management, operational monitoring, and security audit capabilities.

1.5 1.5 References

The following authoritative standards and specifications form the technical basis of this SRS:

[1] ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering. International Organization for Standardization (ISO), 2018. Confirmed current 2024.

[2] HL7 FHIR Release 5 (R5), Fast Healthcare Interoperability Resources Specification. Health Level Seven International, 2023. Official specification: http://hl7.org/fhir/R5/

[3] U.S. Office of the National Coordinator for Health Information Technology (ONC), 21st Century Cures Act Interoperability, Information Blocking, and Health IT Certification, 45 CFR Parts 170 and 171.

[4] OWASP Application Security Verification Standard (ASVS) 5.0.0, OWASP Foundation, May 2025. Standard for web-application technical security controls.

[5] Health Insurance Portability and Accountability Act (HIPAA) Security and Privacy Rules, 45 CFR Part 160 and Part 164 Subparts A, C, and E.

2. 2. Overall Description

2.1 2.1 Product Perspective

DHRMS is a newly engineered, centralized web-based software platform for managing digital healthcare information. It acts as a controlled intermediary between patients, healthcare providers, administrators, compliance auditors, and authorized external healthcare systems.

The platform separates identity management, authorization, clinical data management, consent processing, document management, interoperability, and audit logging into logically distinct subsystems. This separation supports modularity, maintainability, and controlled access across independent operational boundaries.

2.2 2.2 Product Functions

The core functions delivered by DHRMS include:

Subsystem Domain	Major Functional Capabilities
Identity & Authentication	User registration, authentication, session management, logout, and multi-factor verification.
Patient Health Records	Longitudinal patient profile management, demographic storage, and unified record linkage.
Clinical Encounters	Structured documentation of patient-physician interactions, vital signs, and diagnostic assessments.
Diagnostic Management	Diagnostic report creation, metadata tracking, LOINC coding, and clinical file attachment.
Treatment & Prescriptions	Treatment plan documentation, digital prescription authoring, and medication tracking.
Role-Based Access Control	Granular role enforcement, permission verification, and administrative privilege isolation.
Consent Management	Patient-controlled access authorization, scope restriction, temporal expiration, and revocation.
Medical Documents	Secure document upload, MIME-type verification, SHA-256 integrity hashing, and storage.
Audit Trail & Compliance	Immutable chronological logging of access to sensitive records and authorization failures.
Healthcare Interoperability	Structured healthcare data exchange conforming to the HL7 FHIR Release 5 specification.

2.3 2.3 User Classes and Characteristics

User Class	Characteristics & Skill Level	Primary System Functions
Patient	Record owner/subject; normal technical skill expected; web and mobile access.	View longitudinal records, manage profile, grant and revoke provider consent, review access logs.
Healthcare Provider	Authorized medical professional or clinical staff; high domain skill.	Search authorized patients, view clinical histories, record encounters, enter vitals, prescribe medications, add diagnostic reports.
System Administrator	Privileged technical/administrative user; high technical proficiency.	Manage user accounts, configure roles and permissions, maintain system configuration, monitor telemetry.
Compliance / Auditor	Authorized compliance or security personnel; regulatory domain skill.	Review immutable audit records, investigate security events, export regulatory compliance reports.

2.4 2.4 Operating Environment

DHRMS operates as a web-based client-server platform accessible from modern desktop and mobile browsers:

- Client Devices: Desktops, laptops, tablets, and smartphones.
- Supported Browsers: Current Google Chrome, Microsoft Edge, Mozilla Firefox, and Apple Safari releases.
- Server Environment: Containerized Linux (Ubuntu 22.04 LTS / Alpine) utilizing PostgreSQL 16 for relational data, Redis 7 for session caching, and S3-compatible encrypted object storage.
- Networking: TCP/IP with HTTPS (TLS 1.3/1.2) for all protected application communication.

2.5 2.5 Design and Implementation Constraints

- Healthcare information shall be treated as confidential and protected under HIPAA and statutory privacy frameworks.
- Protected resources shall require verified authentication and server-side authorization before access.
- Sensitive communications shall strictly use TLS-protected HTTPS; unencrypted connections are rejected.
- Passwords shall not be stored in plaintext and must be hashed using Argon2id or bcrypt (work factor >= 12).
- Access to medical information shall be permanently and immutably auditable.
- The system shall use standardized healthcare representations conforming to HL7 FHIR Release 5 for interoperability.
- The implementation shall adhere to secure-development practices consistent with OWASP ASVS 5.0.0 (Level 2).

2.6 2.6 User Documentation

The following documentation components will be delivered with the system:

- Patient User Guide: Step-by-step instructions for account access, record review, and consent management.
- Healthcare Provider Clinical Manual: Procedures for encounter charting, e-prescribing, and lab orders.
- Administrator Runbook: User provisioning, role configuration, backup procedures, and security monitoring.
- System Installation and Deployment Guide: Container orchestration, configuration parameters, and database setup.
- API Documentation: OpenAPI and FHIR endpoint specifications for external integrations.

2.7 2.7 Assumptions and Dependencies

Assumptions:

- Users possess internet-enabled computing devices with compatible web browsers.
- Users provide valid registration information during account onboarding.
- Healthcare providers are verified and licensed by an accredited medical body before clinical privileges are enabled.

Dependencies:

- Reliable relational database service (PostgreSQL 16).
- Secure cloud or on-premise object storage service (S3/MinIO).
- Transactional email service (SendGrid / AWS SES) for MFA and notifications.
- External hospital and laboratory gateways where interoperability synchronization is enabled.

3. 3. External Interface Requirements

3.1 3.1 User Interfaces

DHRMS provides role-specific web dashboards. The interface clearly identifies the authenticated user and restricts navigation strictly to authorized capabilities:

Patient Interface Components:

- Profile and demographic account management.
- Longitudinal medical history and allergies display.
- Diagnostic reports and laboratory investigations viewer.

- Active prescriptions and historical treatment plans.
- Clinical encounter chronology and consultation summaries.
- Consent management console (grant, view, narrow scope, and revoke access).
- Access history audit log viewer.

Healthcare Provider Interface Components:

- Multi-attribute patient search (National Health ID, Name, Date of Birth).
- Authorized patient longitudinal chart viewer.
- Clinical encounter documentation editor (SOAP format: Subjective, Objective, Assessment, Plan).
- Diagnostic report entry and attachment uploader.
- Treatment plan and electronic prescription builder.
- Medical document management and secure download interface.

Administrator and Auditor Interface Components:

- User account lifecycle management (create, lock, reset, deactivate).
- Role and permission assignment matrix.
- Audit log forensic query interface (filter by Actor, Action, Resource, Date, Status).
- Security event anomaly monitoring and policy configuration.

3.2 3.2 Hardware Interfaces

DHRMS does not require specialized hardware for its core operations. It executes through standard computing devices and network interfaces. Touch input is supported on mobile devices. Standard optical barcode and QR scanners (via USB HID or camera capture) are supported for reading patient wristband IDs and digital health cards.

3.3 3.3 Software Interfaces

Interface Entity	Purpose	Data Exchanged & Protocol
Database Service	Persistent transactional storage	Users, patients, clinical encounters, diagnostic reports, prescriptions, consent directives, audit events (PostgreSQL 16 over TLS).
Authentication Service	Identity verification and session establishment	Credentials, signed JWT tokens, session claims, MFA verification codes.
Object File Storage	Secure storage of clinical files and scans	Clinical document files (PDF, JPEG, DICOM), SHA-256 hashes, pre-signed temporary download URLs (S3/MinIO API).
Healthcare Interoperability Gateway	Exchange of standardized healthcare data	HL7 FHIR Release 5 JSON resources: Patient, Encounter, Observation, DiagnosticReport over RESTful HTTPS.
Transactional Email Service	Notification and alert delivery	MFA verification OTPs, account activation links, consent revocation alert emails (SMTP over TLS).

3.4 3.4 Communications Interfaces

- All protected browser-server communication SHALL use HTTPS over TLS 1.3 or TLS 1.2 with HSTS enforced.
- REST-style web services SHALL use UTF-8 encoded JSON for request and response payloads.

• Authentication and authorization claims SHALL be transmitted using cryptographically signed Bearer tokens (JWT).
• Medical information SHALL NOT be transmitted across unencrypted network channels.
• External interoperability communication SHALL use secure, authenticated channels (mTLS or OAuth2 Bearer tokens).
• Communication failures SHALL return controlled errors conforming to RFC 7807 without exposing internal implementation details.

4. 4. System Features

4.1 4.1 Identity and Authentication

4.1.1 4.1.1 Description and Priority

The feature provides user registration, authentication, session management, logout, and identity verification. Priority: High

4.1.2 4.1.2 Stimulus/Response Sequences

User submits valid credentials -> system verifies identity -> system establishes authenticated session -> system loads role-specific permissions.

Invalid credentials -> authentication is rejected -> user receives a generic failure message -> security event is logged -> account locked after 5 consecutive failures.

4.1.3 4.1.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-AUTH-001	The system SHALL allow registered users to authenticate using verified credentials.	High	Test
REQ-AUTH-002	The system SHALL validate credentials before granting access to protected application resources.	High	Test
REQ-AUTH-003	Each authenticated session SHALL be uniquely associated with a verified user identity.	High	Inspection
REQ-AUTH-004	The system SHALL provide secure logout that explicitly invalidates active server-side session tokens.	High	Test
REQ-AUTH-005	Unauthenticated users SHALL be denied access to protected resources with HTTP 401 Unauthorized status.	High	Test
REQ-AUTH-006	Passwords SHALL NOT be stored in plaintext and SHALL be hashed using Argon2id with unique salts.	High	Inspection
REQ-AUTH-007	The system SHALL handle invalid credentials without revealing whether a specific account exists.	Medium	Test
REQ-AUTH-008	The system SHOULD support Time-based One-Time Password (TOTP) Multi-Factor Authentication (MFA).	Medium	Demonstration

4.2 4.2 Patient Health Record Management

4.2.1 4.2.1 Description and Priority

The feature manages the creation, storage, retrieval, and controlled modification of longitudinal patient health information. Priority: High

4.2.2 4.2.2 Stimulus/Response Sequences

Authorized user requests record -> authentication is checked -> role and applicable consent are checked -> authorized information is returned -> access is recorded in the audit trail.

Provider requests unconsented record -> consent check fails -> HTTP 403 Forbidden is returned -> security access warning is recorded.

4.2.3 4.2.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-REC-001	The system SHALL maintain a unique National Health Identifier (NHID/UUID) for each patient.	High	Inspection
REQ-REC-002	The system SHALL store patient demographic and profile information (Name, DOB, Gender, Emergency Contact).	High	Test
REQ-REC-003	The system SHALL support longitudinal medical-history information (chronic diseases, allergies, surgical history).	High	Test
REQ-REC-004	The system SHALL support clinical encounter records linked to the longitudinal patient profile.	High	Demonstration
REQ-REC-005	The system SHALL associate health information strictly with the verified patient identity.	High	Test
REQ-REC-006	The system SHALL restrict record access to authorized users and actively consented healthcare providers.	High	Test
REQ-REC-007	Access to sensitive patient records SHALL generate an immutable audit log entry.	High	Test
REQ-REC-008	The system SHALL preserve relational data integrity and ACID properties during record updates.	Medium	Analysis

4.3 4.3 Diagnostic Report Management

4.3.1 4.3.1 Description and Priority

Authorized healthcare providers can create, upload, retrieve, and manage diagnostic reports and lab results. Priority: High

4.3.2 4.3.2 Stimulus/Response Sequences

Provider submits report -> system verifies identity, role, patient association and authorization -> report is stored -> operation is audited -> confirmation is displayed.

Consented provider queries lab results -> metadata validated -> test results displayed -> retrieval audited.

4.3.3 4.3.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-DIAG-001	Authorized healthcare providers SHALL be able to add diagnostic laboratory reports.	High	Test
REQ-DIAG-002	Every diagnostic report SHALL be associated with a valid patient ID and attending provider ID.	High	Inspection
REQ-DIAG-003	The system SHALL store diagnostic metadata including test type, LOINC code, findings, and reference ranges.	Medium	Test
REQ-DIAG-004	Authorized users SHALL be able to retrieve diagnostic reports permitted under active consent scope.	High	Test
REQ-DIAG-005	Diagnostic-report access and retrieval operations SHALL be permanently recorded in the audit trail.	High	Test
REQ-DIAG-006	Unauthorized access attempts to diagnostic reports SHALL be rejected and logged as security events.	High	Test

4.4 4.4 Role-Based Access Control

4.4.1 4.4.1 Description and Priority

The system enforces permissions based on the authenticated user's assigned role and privilege matrix. Priority: High

4.4.2 4.4.2 Stimulus/Response Sequences

User requests protected action -> system identifies role -> system checks permission -> if permitted, operation continues; otherwise, access is denied (HTTP 403) and the security event is recorded.

4.4.3 4.4.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-RBAC-001	Every authenticated user SHALL have at least one assigned system role (Patient, Provider, Admin, Auditor).	High	Inspection
REQ-RBAC-002	The system SHALL verify permissions server-side before executing any protected operation.	High	Test
REQ-RBAC-003	Patients SHALL receive patient-specific permissions restricted to their personal health information.	High	Test
REQ-RBAC-004	Healthcare providers SHALL receive provider-specific permissions restricted to actively consented patients.	High	Test
REQ-RBAC-005	Administrators SHALL receive administrative permissions strictly isolated from clinical medical data.	High	Inspection
REQ-RBAC-006	Users SHALL NOT be permitted to access functions outside their explicitly assigned privileges.	High	Test
REQ-RBAC-007	Modifications to user roles and permission sets SHALL be strictly auditable.	High	Test

4.5 4.5 Consent Lifecycle Management

4.5.1 4.5.1 Description and Priority

The feature enables patients to grant, review, modify, and revoke access permissions for specified healthcare providers and clinical information scopes. Priority: High

4.5.2 4.5.2 Stimulus/Response Sequences

Patient grants consent -> system records scope, authorized provider, and validity -> provider request is evaluated against active consent -> access allowed only when authorization is valid.

Patient revokes consent -> system records revocation -> future requests covered by revoked consent are immediately denied.

4.5.3 4.5.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-CONSENT-001	The system SHALL allow a patient to grant explicit access consent to a registered healthcare provider.	High	Test
REQ-CONSENT-002	The system SHALL allow a patient to revoke access consent at any time with immediate effect.	High	Test
REQ-CONSENT-003	Every consent directive SHALL uniquely identify the authorized party and hospital affiliation.	High	Inspection
REQ-CONSENT-004	Consent directives SHALL specify their applicable data/access scope (e.g., ALL, DIAGNOSTIC, PRESCRIPTIONS).	High	Test
REQ-CONSENT-005	Restricted health data access SHALL be dynamically evaluated against applicable active consent directives.	High	Test
REQ-CONSENT-006	Revoked consent directives SHALL prevent all future provider queries covered by that directive.	High	Test
REQ-CONSENT-007	Consent creation, scope modification, and revocation operations SHALL be logged in the audit trail.	High	Test
REQ-CONSENT-008	Consent records SHOULD support automated expiration based on effective ISO 8601 validity timestamps.	Medium	Demonstration

4.6 4.6 Clinical Encounter Management

4.6.1 4.6.1 Description and Priority

The feature maintains structured records of clinical interactions between patients and certified healthcare providers. Priority: High

4.6.2 4.6.2 Stimulus/Response Sequences

Provider selects authorized patient -> enters encounter information (vitals, chief complaint, ICD-10 diagnosis) -> system validates data -> encounter stored and associated with patient -> access/change audited.

4.6.3 4.6.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-ENC-001	Authorized healthcare providers SHALL be able to create structured clinical encounter records.	High	Test
REQ-ENC-002	Each encounter SHALL be permanently associated with a valid patient ID and attending provider ID.	High	Inspection
REQ-ENC-003	The system SHALL record encounter timestamp, encounter type (Inpatient, Outpatient), and clinical observations.	High	Test
REQ-ENC-004	Authorized users SHALL be able to retrieve permitted encounter records within active consent scope.	High	Test
REQ-ENC-005	All encounter creation, lookup, and modification operations SHALL be auditable.	High	Test

4.7 4.7 Treatment and Prescription Management

4.7.1 4.7.1 Description and Priority

The feature manages therapeutic treatment plans and prescription records authored by certified healthcare providers. Priority: High

4.7.2 4.7.2 Stimulus/Response Sequences

Doctor selects medication, dosage, frequency, and duration -> system validates input -> prescription saved -> patient receives digital record -> operation audited.

4.7.3 4.7.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-TREAT-001	Authorized healthcare providers SHALL be able to record structured treatment and therapy plans.	High	Test
REQ-TREAT-002	Authorized healthcare providers SHALL be able to create structured electronic prescription records.	High	Test
REQ-TREAT-003	Treatment and prescription information SHALL be associated with the correct patient and encounter.	High	Inspection
REQ-TREAT-004	Access to prescriptions SHALL be restricted according to authorization role and patient consent policy.	High	Test
REQ-TREAT-005	Prescription creation, modification, and dispensation queries SHALL be recorded in the audit trail.	High	Test

4.8 4.8 Medical Document Management

4.8.1 4.8.1 Description and Priority

The feature provides controlled upload, storage, retrieval, and management of electronic clinical documents. Priority: Medium

4.8.2 4.8.2 Stimulus/Response Sequences

Authorized user uploads file -> MIME type validated -> SHA-256 computed -> document stored in encrypted storage -> metadata persisted in database -> operation audited.

4.8.3 4.8.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-DOC-001	Authorized users SHALL be able to upload permitted medical documents (PDF, JPEG, PNG, DICOM).	Medium	Test
REQ-DOC-002	Each document SHALL be permanently associated with the relevant patient record and uploader metadata.	High	Inspection
REQ-DOC-003	The system SHALL validate supported document file types and verify SHA-256 checksums upon upload.	High	Test
REQ-DOC-004	The system SHALL enforce configured file-size limits (maximum 25 MB per document).	Medium	Test
REQ-DOC-005	Unauthorized or unconsented users SHALL NOT be permitted to retrieve restricted documents.	High	Test
REQ-DOC-006	All document upload, download, and deletion events SHALL be recorded in the audit trail.	High	Test

4.9 4.9 Audit Trail and Compliance Monitoring

4.9.1 4.9.1 Description and Priority

The feature records security-sensitive and medical-record operations for forensic accountability and compliance review. Priority: High

4.9.2 4.9.2 Stimulus/Response Sequences

Sensitive operation occurs -> system records actor, action, resource, timestamp, IP address, and outcome -> authorized audit personnel review resulting event.

4.9.3 4.9.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-AUDIT-001	The system SHALL record every access, query, modification, and deletion of sensitive patient information.	High	Test
REQ-AUDIT-002	Every audit entry SHALL identify the acting user ID, role, client IP address, and execution timestamp.	High	Inspection
REQ-AUDIT-003	The audit entry SHALL specify the exact operation performed (e.g., READ, WRITE, EXPORT, CONSENT_REVOKE).	High	Test
REQ-AUDIT-004	The audit entry SHALL identify the specific clinical resource URI affected by the operation.	High	Test
REQ-AUDIT-005	Authorization failures and anomalous access attempts SHALL be recorded in the audit trail.	High	Test

Requirement ID	Requirement Statement	Priority	Verification
REQ-AUDIT-006	Authorized compliance personnel SHALL be able to search and export audit records.	High	Demonstration
REQ-AUDIT-007	Audit records SHALL be write-once, append-only, and protected against unauthorized modification or deletion.	High	Analysis

4.10 4.10 Healthcare Data Interoperability

4.10.1 4.10.1 Description and Priority

The system supports structured healthcare information exchange with authorized external systems conforming to HL7 FHIR Release 5 [2]. Priority: Medium

4.10.2 4.10.2 Stimulus/Response Sequences

External hospital system submits authenticated FHIR bundle -> system validates payload against schema -> clinical data ingested -> operation audited -> response returned.

4.10.3 4.10.3 Functional Requirements

Requirement ID	Requirement Statement	Priority	Verification
REQ-INT-001	The system SHOULD support standardized healthcare data representations for clinical data interchange.	Medium	Test
REQ-INT-002	The system SHOULD support FHIR-compatible resources (Patient, Encounter, Observation, DiagnosticReport).	Medium	Test
REQ-INT-003	Imported external healthcare data SHALL be validated and sanitized prior to relational persistence.	High	Test
REQ-INT-004	Interoperability communication SHALL use secure, authenticated channels (mTLS or OAuth2 Bearer tokens).	High	Inspection
REQ-INT-005	All interoperability data ingestions and exports SHALL be recorded in the audit trail.	High	Test

5. 5. Other Nonfunctional Requirements

5.1 5.1 Performance Requirements

The system performance metrics are established to support hospital operations under anticipated peak load:

Requirement ID	Performance Benchmark Statement	Target Metric
REQ-PERF-001	Normal user-interface page rendering SHOULD complete within 2 seconds under expected load.	<= 2.0 s (p95)
REQ-PERF-002	Authentication, credential hashing, and JWT token issuance SHOULD normally complete within 1.5 seconds.	<= 1.5 s (p95)

Requirement ID	Performance Benchmark Statement	Target Metric
REQ-PERF-003	Standard patient-record retrieval and consent evaluation SHOULD normally complete within 1.8 seconds.	<= 1.8 s (p95)
REQ-PERF-004	Concurrent multi-user access SHALL NOT result in unauthorized data exposure, race conditions, or corruption.	500 active users
REQ-PERF-005	Medical document uploads SHALL provide controlled progress tracking and error handling.	>= 10 MB/s throughput
REQ-PERF-006	Performance and load testing SHALL be executed and validated prior to production deployment.	Pass >= 150% peak load

5.2 5.2 Safety Requirements

Requirement ID	Safety Assurance Statement & Safeguard
REQ-SAFE-001	The system SHALL prevent unauthorized or accidental modification of clinical records by enforcing immutable versioning.
REQ-SAFE-002	The system SHALL preserve the transactional integrity (ACID) of stored health information during network or power failures.
REQ-SAFE-003	Destructive or critical clinical operations (e.g., prescribing medications, revoking consent) SHOULD require explicit user confirmation.
REQ-SAFE-004	Critical clinical information SHOULD be backed up regularly in accordance with an approved automated backup schedule.
REQ-SAFE-005	Disaster recovery and restoration procedures SHOULD be periodically tested to verify data recoverability.
REQ-SAFE-006	The user interface SHALL clearly distinguish unavailable or unconsented information from successfully retrieved information.

5.3 5.3 Security Requirements

The security requirements are aligned with the OWASP Application Security Verification Standard (ASVS) 5.0.0 [4] and HIPAA [5]:

Requirement ID	Security Control Specification & Mechanism
REQ-SEC-001	Protected resources SHALL require verified authentication on every inbound request; anonymous access is rejected.
REQ-SEC-002	Server-side authorization SHALL be verified on every protected API endpoint before sensitive operations execute.
REQ-SEC-003	All network communications SHALL use HTTPS over TLS 1.3/1.2; HSTS with max-age=31536000 SHALL be enforced.
REQ-SEC-004	Passwords SHALL be salted and securely hashed using Argon2id and SHALL NOT be stored in plaintext.
REQ-SEC-005	Application sessions SHALL expire after 15 minutes of inactivity; maximum session duration SHALL NOT exceed 8 hours.

Requirement ID	Security Control Specification & Mechanism
REQ-SEC-006	Application inputs SHALL be strictly validated, typed, and parameterized to prevent SQL injection and cross-site scripting (XSS).
REQ-SEC-007	Sensitive stored data at rest (PostgreSQL tablespaces and object storage) SHALL be encrypted using AES-256 with managed KMS keys.

5.4 5.4 Software Quality Attributes

Quality Attribute	Architectural Strategy & Acceptance Metric
Availability	The system SHOULD achieve a 99.9% uptime target during operational service hours (<= 43.8 minutes downtime/month).
Maintainability	The software SHOULD utilize decoupled service layers with standardized RESTful API contracts and modular code organization.
Usability	Users SHOULD be able to complete core clinical tasks with clear navigation, input guidance, and WCAG 2.1 AA accessibility.
Scalability	The architecture SHOULD permit growth in users, records, and document storage through containerized horizontal scaling.
Interoperability	The system SHOULD support standardized healthcare data exchange via HL7 FHIR Release 5 RESTful JSON payloads.
Testability	Requirements SHOULD be verifiable through automated unit, integration, security, and end-to-end regression test suites.

5.5 5.5 Business Rules

Rule ID	Authoritative Business Rule & Policy Enforcement Logic
BR-01	A user SHALL authenticate with valid credentials before accessing any protected system resource.
BR-02	A healthcare provider SHALL only access patient health information for which active patient consent exists.
BR-03	Revoked consent directives SHALL immediately prevent future provider queries covered by the revoked directive.
BR-04	System Administrators SHALL NOT access patient clinical records, diagnoses, or notes directly.
BR-05	Authorization SHALL be strictly enforced server-side and SHALL NOT depend solely on client-side interface controls.

6. 6. Other Requirements

6.1 6.1 Database Requirements

The database shall maintain consistent relationships among users, patients, providers, roles, medical records, encounters, reports, treatment records, prescriptions, consent records, documents, and audit events. The database schema shall be normalized to Third Normal Form (3NF), with referential

integrity strictly maintained via foreign keys. B-tree indexes shall be established on primary lookup columns, and JSONB GIN indexing shall be enabled for HL7 FHIR clinical resources.

6.2 6.2 Data Backup and Recovery

- Regular automated database backups SHALL be executed daily at 02:00 UTC with continuous Write-Ahead-Log (WAL) archiving.
- Backup files SHALL be encrypted at rest (AES-256) and replicated to an off-site cloud storage location.
- The target Recovery Point Objective (RPO) is <= 1.0 hour.
- The target Recovery Time Objective (RTO) is <= 4.0 hours.

6.3 6.3 Data Privacy

Patient health information shall be treated as confidential Protected Health Information (PHI). DHRMS shall implement tenant isolation, consent verification on every clinical read, and audit logging for all PHI disclosures in conformance with HIPAA [5] and statutory privacy mandates.

6.4 6.4 Data Retention

Clinical medical records, diagnostic reports, and encounter summaries shall be retained for a mandatory minimum period of 7 years in compliance with standard healthcare record retention laws. Security audit trail records shall be retained permanently in immutable cold storage for forensic traceability.

6.5 6.5 Internationalization

English shall be the primary operational language for Version 2.0. The application architecture shall utilize full UTF-8 character encoding to support international medical nomenclature. Timestamps shall be stored in ISO 8601 UTC format with localized client-side timezone presentation.

6.6 6.6 Enterprise Logging

The application shall generate structured JSON log streams capturing authentication events, authorization failures, sensitive record access, consent lifecycle events, administrative operations, and unhandled system exceptions with unique correlation IDs.

7. Appendix A: Glossary

This glossary defines all domain-specific and technical terms utilized throughout this SRS:

Term / Acronym	Authoritative Definition & Context
Authentication	The verification process confirming that an entity requesting system access possesses the claimed identity through credentials or tokens.
Authorization	The determination process verifying whether an authenticated user is permitted by role and consent policy to perform a specific operation.
Audit Trail	A chronological, tamper-evident record of system activities documenting who performed what action, on which resource, at what time.
Consent Directive	A formal authorization record authored by a patient granting, modifying, or revoking provider access to health information.

Term / Acronym	Authoritative Definition & Context
Clinical Encounter	A documented interaction between a patient and certified healthcare provider containing clinical notes, vitals, and diagnoses.
Diagnostic Report	A clinical document summarizing laboratory, imaging, or pathology investigation results associated with a patient.
Electronic Health Record (EHR)	A longitudinal digital collection of patient health information spanning encounters, medications, diagnoses, and lab results.
HL7 FHIR R5	Fast Healthcare Interoperability Resources (Release 5): An international standard for healthcare data exchange using modular RESTful JSON resources.
HIPAA	Health Insurance Portability and Accountability Act: Statutory benchmark governing the confidentiality and security of Protected Health Information.
Healthcare Provider	An authenticated, licensed medical professional or healthcare organization authorized to access and document health records.
JWT (JSON Web Token)	A compact, URL-safe means of representing signed claims securely transmitted between client and server (RFC 7519).
LOINC	Logical Observation Identifiers Names and Codes: Universal standard coding system for identifying laboratory test observations.
Patient	The individual whose personal healthcare records and medical history are managed and protected within DHRMS.
Protected Health Information (PHI)	Individually identifiable health information relating to an individual's medical condition, clinical history, or healthcare provision.
RBAC	Role-Based Access Control: An access-control mechanism where authorization decisions are governed strictly by assigned user roles.
RPO / RTO	Recovery Point Objective (maximum acceptable data loss) and Recovery Time Objective (maximum acceptable recovery duration).
TLS (Transport Layer Security)	Cryptographic protocol providing communications security, confidentiality, and data integrity over computer networks.
API	Application Programming Interface: Standardized interface enabling communication between client frontends and backend services.

8. Appendix B: Analysis Models

In accordance with ISO/IEC/IEEE 29148:2018, this appendix provides structured textual and tabular analysis models defining system boundaries, use case specifications, relational data dictionaries, access-control algorithms, state-transition matrices, and requirement traceability.

8.1 B.1 System Context Model and Interface Exchange Matrix

The following matrix specifies the data exchange boundaries between external actors, systems, and DHRMS:

External Entity	Direction	Information Exchanged	Protocol / Channel	Security Boundary
Patient	Inbound	Login credentials, Profile updates, Consent grant/revoke forms	HTTPS / JSON	Argon2id + JWT Auth
Patient	Outbound	Health summary, Prescriptions, Lab reports, Access history	HTTPS / JSON	Patient-owner scope check
Healthcare Provider	Inbound	Credentials, SOAP clinical notes, Prescriptions, Lab orders	HTTPS / JSON	Role check + Active Patient Consent
Healthcare Provider	Outbound	Consented patient records, Diagnostic reports, Encounter history	HTTPS / JSON	Scoped authorization validation
System Administrator	Inbound	User onboarding, Role mappings, System configuration	HTTPS / JSON	Super-Admin privilege verification
Compliance Auditor	Outbound	Tamper-evident audit logs, Forensic event query results	HTTPS / JSON	Audit read-only privilege
External Lab / Hospital	Bi-directional	HL7 FHIR R5 resources (Patient, Observation, DiagnosticReport)	REST / HTTPS	Mutual TLS / OAuth2 Bearer token

8.2 B.2 Detailed Use Case Specifications

Use Case UC-01: User Authentication and Multi-Factor Verification

Element	Specification Details
Use Case ID	UC-01
Actor(s)	Patient, Healthcare Provider, System Administrator, Compliance Auditor
Preconditions	User possesses a registered, active account in DHRMS.
Main Flow	1. User navigates to portal and submits username/password.2. System retrieves credential hash and verifies identity using Argon2id.3. If MFA enabled, system prompts for TOTP 6-digit code.4. User provides valid TOTP code; system establishes authenticated session.5. System issues cryptographically signed JWT token with role claims.6. System redirects user to role-specific dashboard.
Alternative Flows	2a. Invalid credentials -> Error displayed; failed login logged; session blocked.3a. Invalid MFA code -> Second attempt allowed; after 3 failures, session aborted.
Postconditions	User is authenticated with a time-limited session; audit log records successful login.

Use Case UC-02: Longitudinal Patient Record Retrieval

Element	Specification Details
Use Case ID	UC-02
Actor(s)	Healthcare Provider, Patient
Preconditions	Provider is authenticated; patient possesses registered health profile.

Element	Specification Details
Main Flow	1. Provider searches patient by National Health ID, Name, or DOB.2. System evaluates Provider role permissions.3. System checks for an active, unexpired Consent Directive granted by the patient.4. If valid consent exists, system retrieves longitudinal records, encounters, and vitals.5. Clinical chart is rendered on provider workstation.6. System appends read event to immutable audit log.
Alternative Flows	3a. No active consent exists -> System rejects request with HTTP 403 Consent Denied; access failure recorded in audit log.
Postconditions	Authorized records displayed; audit log updated with viewing timestamp.

Use Case UC-03: Patient Consent Lifecycle Management

Element	Specification Details
Use Case ID	UC-03
Actor(s)	Patient
Preconditions	Patient is authenticated on Patient Portal.
Main Flow	1. Patient opens Consent Management console.2. Patient clicks 'Grant Consent' and selects provider, access scope, and validity duration.3. Patient confirms authorization directive.4. System creates Consent Record in database with status ACTIVE.5. System logs consent grant in audit trail.6. System displays active directive in patient's consent list.
Alternative Flows	2a. Patient clicks 'Revoke' on an active directive -> System prompts for confirmation -> Status set to REVOKED -> Future provider queries immediately blocked.
Postconditions	Consent directive state updated in database; audit trail permanently recorded.

8.3 B.3 Logical Data Entities and Relational Schema Dictionary

The following data dictionary defines the core logical entities and relational schema attributes:

Entity Name	Primary Key	Foreign Keys	Key Attributes	Relational Description
USERS	user_id (UUID)	role_id (ROLES)	username, email, password_hash, mfa_secret, status	Root authentication entity; 1:1 relation to Patient or Provider.
ROLES	role_id (INT)	None	role_name, permissions_mask, description	Defines system privilege boundaries (Patient, Provider, Admin, Auditor).
PATIENTS	patient_id (UUID)	user_id (USERS)	national_health_id, full_name, dob, gender, blood_type	Patient profile holding demographics; parent of clinical records.
PROVIDERS	provider_id (UUID)	user_id (USERS)	license_number, specialization, department, hospital_id	Certified healthcare provider clinical profile.

Entity Name	Primary Key	Foreign Keys	Key Attributes	Relational Description
CLINICAL_ENCOUNTERS	encounter_id (UUID)	patient_id, provider_id	encounter_date, encounter_type, chief_complaint, vitals, notes	Documented patient-doctor consultations (SOAP format).
DIAGNOSTIC_REPORTS	report_id (UUID)	encounter_id, patient_id	test_code_loinc, findings_text, result_value, unit, lab_date	Structured laboratory and diagnostic test observations.
PRESCRIPTIONS	prescription_id (UUID)	encounter_id, patient_id	medication_name, dosage, frequency, route, duration, status	Electronic prescription records authored by attending providers.
MEDICAL_DOCUMENTS	document_id (UUID)	patient_id, uploader_id	file_name, storage_path_s3, mime_type, file_size, sha256_hash	Binary medical document attachments stored in S3/MinIO.
CONSENT_RECORDS	consent_id (UUID)	patient_id, provider_id	access_scope, valid_from, valid_until, status (ACTIVE/REVOKED)	Dynamic authorization directives governing clinical access.
AUDIT_EVENTS	audit_id (BIGINT)	actor_user_id (USERS)	action_type, resource_uri, outcome, client_ip, timestamp, hash	Immutable chronological security log for compliance.

8.4 B.4 Access-Control Flow and Verification Algorithm

Every protected inbound API request is processed through the following multi-stage verification pipeline:

Algorithm 1: Server-Side Authorization Pipeline

1. Inbound Request Received: API Gateway intercepts HTTP request.2. Authentication Verification: IF Authorization Bearer token is missing OR invalid THEN Return HTTP 401 Unauthorized; Log Security Event; TERMINATE. Extract user_id and assigned role from validated token claims.3. RBAC Policy Check: IF role does NOT possess required endpoint permission THEN Return HTTP 403 Forbidden; Log RBAC Violation; TERMINATE.4. Target Resource Classification: IF target resource does NOT contain Protected Health Information (PHI) THEN Execute Service Operation; Log Audit Event; Return HTTP 200 OK; TERMINATE.5. Patient Consent Directive Verification: Query active CONSENT_RECORDS where patient_id = target_patient_id AND provider_id = acting_provider_id. IF no directive exists OR status != 'ACTIVE' OR current_time > valid_until THEN Return HTTP 403 Consent Denied; Log Consent Violation; TERMINATE. IF requested data category NOT in consent access_scope THEN Return HTTP 403 Scope Restricted; Log Scope Violation; TERMINATE.6. Execution & Persistence: Execute clinical database query within ACID transaction boundary.7. Immutable Audit Vault Logging: Compute SHA-256 hash of event details (actor, action, resource, timestamp). Append audit record to AUDIT_EVENTS table.8. Response Dispatch: Encrypt response payload via TLS and return HTTP 200 OK.

8.5 B.5 Consent Lifecycle State-Transition Matrix

The following state machine governs patient authorization directives from inception to termination:

Current State	Trigger Event	Guard Condition	Next State	Resulting System Action
NO_CONSENT	grantConsent()	Valid provider and scope specified	ACTIVE	Consent record created; provider queries permitted within scope.
ACTIVE	modifyScope()	Patient authenticated as record owner	ACTIVE	Scope updated; audit event logged; new boundary enforced.
ACTIVE	revokeConsent()	Patient triggers manual revocation	REVOKED	Status updated immediately; active provider tokens invalidated.
ACTIVE	timerExpiry()	Current time > valid_until timestamp	EXPIRED	Status updated to EXPIRED; provider access automatically denied.
REVOKED	readRecord()	Provider attempts clinical query	REVOKED	Access denied (HTTP 403); security audit event logged.
EXPIRED	readRecord()	Provider attempts clinical query	EXPIRED	Access denied (HTTP 403); notification sent to renew consent.

8.6 B.6 Functional Subsystem Traceability Matrix

The traceability matrix maps business rules and use cases to verified functional requirement identifiers:

Business Rule	Primary Use Case	Functional Requirements Covered	Verification Strategy
BR-01 (Mandatory Auth)	UC-01 (Auth & MFA)	REQ-AUTH-001 through REQ-AUTH-008	Automated Security Test Suite
BR-02 (Consent Sovereignty)	UC-03 (Consent Mgmt)	REQ-CONSENT-001 through REQ-CONSENT-008	Integration Test & Simulation
BR-03 (Immediate Revocation)	UC-03 (Consent Mgmt)	REQ-CONSENT-002, REQ-CONSENT-006	End-to-End Boundary Test
BR-04 (Separation of Duties)	UC-01, UC-02	REQ-RBAC-001 through REQ-RBAC-007	Role Privilege Penetration Test
BR-05 (Server-Side Enforce)	All Clinical UCs	REQ-REC-006, REQ-DIAG-006, REQ-DOC-005	API Gateway Fuzzing & Static Analysis

9. Appendix C: To Be Determined List

All 15 preliminary 'To Be Determined' (TBD) architectural parameters identified during initial project scoping have been formally evaluated, sized, and resolved by the engineering team:

ID	Architectural Parameter	Authoritative Engineering Resolution	Rationale & Standard Metric
TBD-001	Final database technology	PostgreSQL 16 Relational Engine	ACID compliance, JSONB support for FHIR, robust row-level security.
TBD-002	Hosting environment	Docker / Kubernetes Container Cluster	Portable deployment across cloud and institutional on-premise servers.
TBD-003	Backup frequency	Automated Daily at 02:00 UTC + WAL	Continuous write-ahead logging prevents transaction loss during failures.
TBD-004	Recovery Point (RPO)	RPO <= 1.0 Hour	Healthcare continuity standard for minimal transactional data loss.
TBD-005	Recovery Time (RTO)	RTO <= 4.0 Hours	Automated container orchestration restores failed services rapidly.
TBD-006	Data retention periods	7 Years Clinical / Permanent Audit	Complies with statutory clinical retention mandates; audit records kept permanently.
TBD-007	Regulatory framework	HIPAA & National Healthcare Guidelines	Standard security benchmark for Protected Health Information.
TBD-008	Final MFA mechanism	TOTP (RFC 6238) Authenticator App	Secure offline code generation avoiding vulnerable SMS SS7 interception.
TBD-009	External integrations	Standardized LIS & Hospital EMR	RESTful JSON data exchange for lab orders and imaging summaries.
TBD-010	FHIR resources scope	Patient, Encounter, Observation, Rx	Covers core clinical workflows specified in HL7 FHIR Release 5 US Core / International Profiles.
TBD-011	Concurrent users cap	500 Active Users (Burst to 2,500)	Sized for medium-scale multi-department hospital campus operational capacity.
TBD-012	Document upload cap	25.0 MB per Upload Stream	Accommodates high-resolution multi-page diagnostic PDFs and DICOM slices.
TBD-013	Availability target	99.9% Production Uptime (Three 9s)	Allows <= 43.8 minutes planned maintenance downtime per calendar month.
TBD-014	Notification provider	Transactional SMTP (SendGrid/SES)	Queue-backed transactional email delivery with TLS encryption and bounce tracking.
TBD-015	Deployment architecture	3-Tier Micro-Modular Architecture	React SPA client -> NodeJS/Express REST API Gateway -> PostgreSQL + S3 Object Store.

10. Appendix D: Team Contributions and Agile Governance

This project was executed by a collaborative 4-member engineering team under the academic guidance of Mr. Navdesh Singh for the Software Engineering course:

Member Name	Designated Role	Roll Number	Subsystem Ownership & Contributions
Abhinav Singh	Project Leader & Architect	202401100200014	Overall system architecture, ISO/IEC/IEEE 29148 alignment, Context Model, RBAC security design, and team coordination.
Abhinav Pathak	Lead Full-Stack Developer	202401100200013	Relational database schema modeling, PostgreSQL entity design, FHIR R5 API contracts, and Medical Document Management (4.8).
Abhishek Kumar Verma	Scrum Master & Analyst	202401100200015	Sprint backlog management, functional requirements authoring (4.1-4.7), and Consent Lifecycle state machine design.
Abhishek Yadav	QA Lead & Tester	202401100200016	Verification criteria specification, OWASP ASVS 5.0 security compliance review, performance benchmarking, and test harness planning.

Agile Sprint Execution Summary:

- Sprint 1 (Weeks 1-2): Stakeholder elicitation, user persona definition, regulatory privacy review, and baseline scope drafting.
- Sprint 2 (Weeks 3-4): Detailed functional requirement specification for Identity, EHR, Encounters, Prescriptions, and Consent rules.
- Sprint 3 (Weeks 5-6): Analysis modeling including Context Matrix, Use Case tables, Relational Schema Dictionary, and State Machines.
- Sprint 4 (Weeks 7-8): Non-functional requirement verification, OWASP ASVS security control mapping, TBD resolution, and faculty review.

Faculty Supervisor Course Review Block:

Evaluation Context	Faculty Assessment & Baseline Status	Faculty Supervisor
Software Engineering Course AssessmentDepartment of CSEAcademic Year: 2026-2027	The Software Requirements Specification adheres to the ISO/IEC/IEEE 29148:2018 and IEEE 830-1998 standards. The functional decomposed requirements, analytical models, and security controls are approved for implementation.	Mr. Navdesh SinghFaculty Guide / SupervisorDepartment of Computer Science and Engineering